

The Signal Door Test

國中 7 年級

第 1 週

預計 66 分鐘

這週要帶走的能力

- 讀懂科技門測試中哪個裝置出了問題。
- 學會依主詞選 am / is / are。
- 掌握 10 個科技解謎核心字。
- 用文章證據寫出簡短測試結論。

先讀文章並在每段旁邊記『測了什麼、看到什麼、所以知道什麼』。做文法題時先圈主詞，再選 am / is / are。閱讀推論題一定回文章找證據。

1 先自己讀 2 圈起生字 3 獨立作答 4 訂正思考 5 AI輔助解惑

先想一想

想像你做了一扇會自動開啟的遊戲基地門，但按鈕亮了、門卻沒開。你第一個會檢查哪個部分？用中文寫一句。

閱讀：The Signal Door Test

情境背景：兩位學生在方塊建造遊戲中設計一扇自動門。門有感應器、開關和訊號燈，但第一次測試失敗。他們不亂拆，而是一次只改一個地方並記錄結果。

讀法提示：每段找一個『觀察到的結果』，不要先猜原因。；看到 working / not correct 時，回頭找哪個測試支持這個判斷。；推論題要分清楚『文章直接說』與『根據多次測試合理推出』。

Kai and Nora build an automatic door in a block-building game. A sensor is in front of the door, and a switch is on the wall. The sensor should detect a player and send a signal through a short route. If the signal reaches the control device, the door opens. Their first test looks strange: the sensor light is green, but the door stays closed.

TEST RULE

Change only one part at a time. Record the result before the next test.

Kai thinks the sensor is broken, but Nora checks the evidence first. She stands near the door three times. Each time, the sensor light turns green. 'The sensor is working,' she says. They then test the wall switch. The switch light is also green, but the control device has no light. Now they know the problem is between the switch and the device, not at the sensor.

They inspect the signal route. Two blocks are connected in the wrong pattern. Kai changes only those two blocks and keeps everything else the same. On the next test, the control device turns green and the door opens. They repeat the test twice. The same result happens both times.

Nora writes a short report: 'The sensor and switch are working. The old signal route is not correct. After we correct the connection pattern, the signal reaches the device and the door opens.' Kai notices something important. They did not fix the door by changing many things quickly. They found the problem because every test answered one clear question.

【學習策略加深】

把除錯變成三欄表

真正的除錯不是一次改很多東西。把每次測試整理成『測試位置 | 看到的結果 | 可以排除什麼』，會更容易找出問題。

延伸小任務

把文章中的 sensor、switch、signal route 三次檢查各整理成一行。

核心單字

signal (n.) The green light sends a signal to the door. 綠燈把訊號送到門。	訊號	device (n.) This device controls the door. 這個裝置控制門。	裝置；設備
connect (v.) Connect the sensor to the switch. 把感應器連接到開關。	連接	switch (n./v.) The switch is beside the door. 開關在門旁邊。	開關；切換

sensor (n.)

感應器

The sensor detects a player near the door.

感應器會偵測門附近的玩家。

detect (v.)

偵測；察覺

The sensor can detect movement.

感應器可以偵測移動。

route (n.)

路線；傳遞路徑

The signal follows a short route.

訊號沿著一條短路徑傳遞。

pattern (n.)

規律；模式

The lights follow the same pattern each time.

燈每次都遵循相同模式。

correct (adj./v.)

正確的；修正

The first connection is not correct.

第一個連接方式不正確。

result (n.)

結果

They record the result after every test.

他們在每次測試後記錄結果。

觀念解說：先看主詞，再選 am / is / are

be 動詞會跟主詞一起變化：I 用 am；he / she / it 與單數名詞用 is；you / we / they 與複數名詞用 are。先找主詞，不要只看句尾。

核心句型結構

I am + ...

核心句型結構

He / She / It is + ...

核心句型結構

You / We / They are + ...

完整示範

The sensor is green.

解析思考：sensor 是單數，相當於 it，所以用 is。

Kai and Nora are careful.

解析思考：兩個人是複數，相當於 they，所以用 are。

I am ready to test the door.

解析思考：主詞 I 固定搭配 am。

容易踩到的盲點與陷阱

✗ 錯誤

~~The lights is green.~~

✓ 正確

The lights are green.

💡 為什麼：lights 是複數，所以用 are。

觀念解說：除錯閱讀：測試 → 結果 → 結論

文章裡每個結論都應該有測試證據。先問『他們測了什麼？』，再找『看到什麼？』，最後才寫『所以知道什麼』。

核心句型結構

Test → Result → Conclusion

核心句型結構

Evidence: ... → Therefore: ...

完整示範

Sensor test → green light → sensor is working

解析思考：感應器連續三次亮綠燈，所以可以先排除感應器完全失效。

Device has no light → problem is before the device

解析思考：開關有亮但控制裝置沒亮，問題位置被縮小到兩者之間。

容易踩到的盲點與陷阱

✗ 錯誤 ~~The door stays closed, so the sensor is broken.~~

✓ 正確 The door stays closed, but the sensor turns green three times, so the sensor is probably working.

💡 為什麼：不能只看最後結果，要把感應器自己的測試證據一起考慮。

觀念解說：用簡單句寫測試報告

先寫狀態，再寫結果。第一週不需要複雜句型，重點是主詞、be 動詞與證據一致。

核心句型結構

The ____ is / are ____.

核心句型結構

The result is ____.

完整示範

The sensor is working.

解析思考：單數 sensor 搭配 is，直接描述測試後狀態。

The two lights are green.

解析思考：two lights 是複數，搭配 are。

容易踩到的盲點與陷阱

✗ 錯誤

The two lights is green.

✓ 正確

The two lights are green.

💡 為什麼：主詞是複數 two lights，要用 are。

跟著示範

先抓主詞、核心字與直接證據

先圈主詞或文章中的關鍵句，再選答案。

提示：文法題先看主詞是單數還是複數。

G1

The control device _____ dark during the second test.

(A) is

(B) are

(C) am

(D) be

G2

In the report, what does route most likely mean?

- (A) the path the signal follows
- (B) the color of the door
- (C) a person who tests games
- (D) the time of the test

G3

What happens every time Nora stands near the door?

- (A) The sensor light turns green
- (B) The control device turns green
- (C) The wall switch breaks
- (D) The door opens immediately

自己完成

自己整理測試證據

先完成證據整理，再回答後面兩題。

I1

Organizer: Write one row for each test: part tested → result → what they learn. Include sensor, switch/device, and signal route.

I2

Why do Kai and Nora repeat the final test twice?

- (A) To check that the same fix gives the same result
- (B) To make the sensor light change color
- (C) To test a completely different door
- (D) To make the report longer

I3

Choose the correct sentence.

- (A) The sensor and switch are working.
- (B) The sensor and switch is working.
- (C) The sensor and switch am working.
- (D) The sensor and switch be working.

會考型轉移

會考式證據判讀

每題都要能指出文章哪個測試或結果支持答案。

提示：不要把『可能原因』和『測試後確認的結論』混在一起。

C1

What is the main lesson of the report?

- (A) Changing one part at a time helps identify a problem
- (B) Automatic doors always need more sensors
- (C) A green light means every device is correct
- (D) Fast builders should change many parts together

C2

Which part is most likely the real problem before the fix?

- (A) The connection pattern in the signal route
- (B) The sensor itself
- (C) Nora's position near the door
- (D) The final report

C3

The text says they inspect the signal route. What does inspect most likely mean here?

- (A) look at something carefully
- (B) build something without a plan
- (C) hide something from view
- (D) turn something off forever

寫出自己的答案

寫出你的測試結論

用完整英文句子，重點是主詞與 be 動詞正確。

P1

Write one sentence about the sensor and one sentence about the two lights. Use is / are correctly.

P2

Write one short test conclusion using at least one core word such as signal, route, pattern, or result.

隔天提取

過一段時間再提取

不要回看文法框或單字表。

R1

The two blocks _____ connected in the wrong pattern.

(A) are

(B) is

(C) am

(D) be

R2

Which core word means 『偵測』?

(A) detect

(B) route

(C) result

(D) switch

自我檢核

完成今日學習後，請勾選已掌握的項目，並寫下心得或問題：

- ☐ 我會先看主詞再選 am / is / are。
 - ☐ 我能把文章中的測試、結果與結論連起來。
 - ☐ 我做推論題時有找到文章證據。
 - ☐ 我能認出本週大部分核心字。
 - ☐ 我能用簡單英文寫出一個測試結果。
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帶走一點，隔天再想一次

任務目標：隔一天確認基礎文法、核心字與證據整理能否在沒有提示時再次使用。(預計 8 分鐘)

建議在完成教材隔天再作答，測試自己是否真正記住。

H1

Choose the correct sentence.

- | | |
|--------------------------|---------------------------|
| (A) The device is ready. | (B) The device are ready. |
| (C) The device am ready. | (D) The device be ready. |

H2

The sensor sends a _____ to the control device.

- | | |
|------------|-------------|
| (A) signal | (B) pattern |
| (C) result | (D) switch |

H3

Give two pieces of evidence showing that the sensor is probably not the broken part.
